SIMATS Online Programming Lab

C PROGRAMMING

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QUESTIONS 25 / 25 completed

Rank

RANK

Write a C program to find the rank of a matrix.

PROGRAM EDITOR

C | Run | Save

```c
#include <stdio.h>

int main() {
    int r1, c1, r2, c2, i, j, k;

    // Read dimensions of the first matrix
    printf("Enter rows and columns of the first matrix: ");
    if (scanf("%d %d", &r1, &c1) != 2 || r1 <= 0 || c1 <= 0) {
        return 1;
    }

    // Read dimensions of the second matrix
    printf("Enter rows and columns of the second matrix: ");
    if (scanf("%d %d", &r2, &c2) != 2 || r2 <= 0 || c2 <= 0) {
        return 1;
    }

    // Condition for matrix multiplication: columns of 1st matrix r
    if (c1 != r2) {
        printf("Matrix multiplication is not possible. Columns of
```

OUTPUT

```
Enter rows and columns of the first matrix: Enter rows and
columns of the second matrix: Matrix multiplication is not
possible. Columns of the first matrix must match rows of the
second matrix.
```

INPUT

```
2 2
1 2
3 4
2 2
```

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